Proceeding of the National Conference on Geometry, Analysis and Fluid Mechanics
@ Government First Grade College, Koppa, Karnataka, INDIA
ISBN: 978-81-926808-1-1

RADIATION EFFECT ON UNSTEADY MHD STRETCHING SHEET IN A MICROPOLAR FLUID WITH SUCTION

E. Manjoolatha
Department of Mathematics, Sri Venkateswara University, Tirupati – 517502.

N. Bhaskar Reddy, S.Suneetha

Assistant Professor, Department of Applied Mathematics, Yogi Vemana University, Kadapa-516003

E. Manjoolatha, N. Bhaskar Reddy and S.Suneetha

Abstract: *The unsteady stretching sheet in a micropolar fluid with heat and mass fluxes in the presence of magnetic field and radiation. The governing partial differential equations are transformed into a system of ordinary differential equations, which is then solved numerically by a fourth order Runge-Kutta method along with shooting technique. Numerical results are obtained for the skin-friction coefficient, the local Nusselt number and Sherwood number as well as the velocity, temperature and concentration profiles for different values of the governing parameters, namely, the magnetic parameter, material parameter, radiation parameter and Prandtl number.*

Keywords : Unsteady flow, heat and mass transfer, stretching sheet, MHD, Micropolar fluid, suction.

1 Introduction

The fluid dynamics due to a stretching surface is important in manufacturing processes. Examples are numerous and they include the aerodynamic extrusion of plastic sheets, the boundary layer along a liquid film in condensation processes, paper production, glass blowing, metal spinning and drawing plastic films. The quality of the final product depends on the rate of heat transfer and mass transfer at the stretching surface. Crane [1] first obtained an elegant analytical solution to the boundary layer equations for the problem of steady two-dimensional flow due to a stretching surface in a quiescent incompressible fluid. Gupta and Gupta [2] extended the problem posed by Crane [1] to a permeable sheet and obtained closed-form solution, while Grubka and